#1 Check Euler's invariants on a graph.
#2 Temporary variables for number of vertices and edges in `graph`.
#3 First constraint: if a graph is planar then $|E| \leq 3|V| - 6$.
#4 Second (stricter) constraint: if a graph is planar and doesn't have any cycle of length 3, then $|E| \leq 2|V| - 4$.

The method to check Euler's constraints, shown in listing 15.3, is directly derived from the formulas in section 15.2.2; the hardest part is verifying that a graph doesn't have any cycle of length 3: this can be done using a modified version of DFS that returns all cycles, and runs in linear $O(V+E)$ time. Since this is quite expensive and requires a non-trivial effort to write and maintain the code, the benefit of including this second check is arguable and – especially on your first try – it might not be worth it: you can start with checking the first constraint only, and remove the if at line #4.

Listing 15.4 Planarity Testing utility methods

```
function
isK5(graph)
      #1
  if |graph.vertices| == 5 and |graph.simpleEdges| == 10
then                  #2
    return true
  return false
```

#1 Method `isK5` takes a graph and checks if it is the complete graph with 5 vertices.